

```
import cv2
import matplotlib.pyplot as plt
import numpy as np
from sklearn.cluster import KMeans

image = cv2.imread("Chhayaassign.jpeg")
if image is None:
    raise FileNotFoundError("Could not find Datasets/ass2.jpeg")
image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

original_shape = image.shape
pixels = image.reshape(-1, 3)

kmeans = KMeans(n_clusters=4, random_state=42, n_init=10)
labels = kmeans.fit_predict(pixels)

centers = np.uint8(kmeans.cluster_centers_)

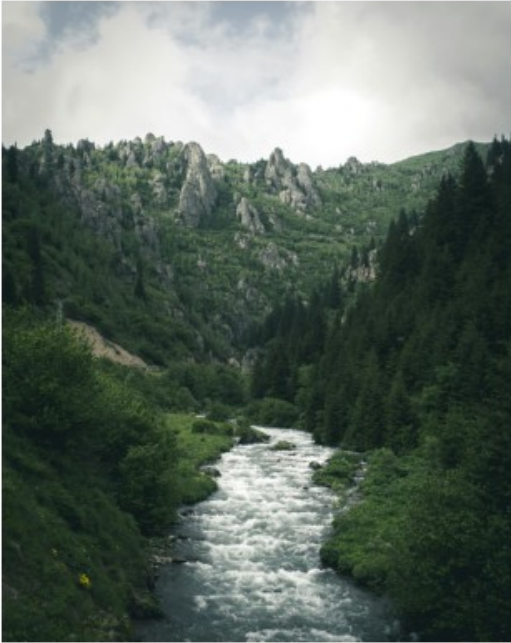
segmented_pixels = centers[labels]
segmented_image = segmented_pixels.reshape(original_shape)

plt.figure(figsize=(10, 5))
plt.subplot(1, 2, 1)
plt.title("Original Image")
plt.imshow(image)
plt.axis("off")

plt.subplot(1, 2, 2)
plt.title("Segmented Image (K=4)")
plt.imshow(segmented_image)
plt.axis("off")

plt.show()
```

Original Image



Segmented Image (K=4)

